σ-Equivariant magmas on AG(2, 3): finite landscape, intrinsic selection, and linearization

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Abstract

We study a finite family of σ-equivariant magmas on

$$M = S \times S, \qquad S = \mathbb{F}_3,$$

with left-row output and a fixed same-row Steiner rule. The normal form is

$$\Omega' \cong S^{18} \times S^3 \cong S^{21}, \qquad |\Omega'| = 3^{21}.$$

The paper has three layers. First, it maps the finite landscape: the certified associativity atlas gives

$$63 \leq \mathrm{Assoc} \leq 597,$$

with endpoint loci and exact partition function, while the global hidden-continuation theorem gives

$$\min H_{tot} = -2268, \qquad \max H_{tot} = 7302, \qquad \max_{N_-=0} H_{tot} = 7020.$$

The layer above the pure frontier consists of twelve impure signed-cancellation maximizers.

Second, the central selection theorem isolates one magma by four internal finite conditions:

$$\boxed{\mathcal{H}_{acc} = 0, \qquad H_{diag} = 0, \qquad \text{diagonal idempotence}, \qquad \text{pure } C/J.}$$

These conditions force the PAB multiplication

$$(r_1, c_1) \cdot (r_2, c_2) = \begin{cases} (r_1, r_2), & r_1 \neq r_2, \\ (r_1, \overline{c_1 c_2}), & r_1 = r_2,\ c_1 \neq c_2, \\ (r_1, r_1), & r_1 = r_2,\ c_1 = c_2. \end{cases}$$

Third, after selection, PAB admits a linear/operator passport: rank-three left translations, a finite composition envelope, endomorphism and intertwiner structure, a binary term algebra of size

$$|T_2(\mathrm{PAB})| = 630,$$

a row-contraction matrix bridge, a hidden-continuation operator identity, and companion carrier-level Lie and signed symplectic structures.

The organizing implication is

$$\boxed{\text{finite landscape}} \Longrightarrow \boxed{\text{intrinsic finite selection}} \Longrightarrow \boxed{\text{linearization of the selected magma}}.$$

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0 Introduction

0.1 Aim

The starting point is a finite landscape of multiplications on

$$M = S \times S, \qquad S = \mathbb{F}_3.$$

The first coordinate is a row, the second is a column. The family is defined by left-row output, the same-row Steiner rule, and σ-equivariance. These axioms leave exactly twenty-one ternary coordinates,

$$\Omega' \cong S^{18} \times S^3 \cong S^{21}, \qquad |\Omega'| = 3^{21} = 10\,460\,353\,203.$$

The manuscript follows one spine:

$$\text{finite landscape} \longrightarrow \text{intrinsic finite selection} \longrightarrow \text{linearization after selection.}$$

The global atlases describe the ambient landscape; the compact selector isolates PAB; the operator, matrix-carrier, Lie, symplectic, and term-algebra results describe the selected magma.

0.2 Main result: compact selection

The central theorem is the uniqueness of PAB under four finite conditions:

$$\boxed{\mathcal{H}_{acc} = 0, \qquad H_{diag} = 0, \qquad \text{diagonal idempotence}, \qquad \text{pure } C/J.}$$

Here

$$\mathcal{H}_{acc} = I(C_{out}; C_1, C_2 \mid R_1, R_2)$$

is zero hidden-column access in the cross-row channel, and

$$H_{diag} = -\sum_s p_s \log_2 p_s$$

is the Shannon entropy of the same-point diagonal output multiset. Their zero sets are the finite structural conditions of column-blindness and diagonal constancy. Diagonal idempotence fixes the constant diagonal to $d = 000$. The final pure C/J condition is imposed on the six directed edges

$$M^{\times} = \{(r, c) : r \neq c\}.$$

The selected multiplication is

$$(r_1, c_1) \cdot (r_2, c_2) = \begin{cases} (r_1, r_2), & r_1 \neq r_2, \\ (r_1, \overline{c_1 c_2}), & r_1 = r_2,\ c_1 \neq c_2, \\ (r_1, r_1), & r_1 = r_2,\ c_1 = c_2. \end{cases}$$

A longer explanatory chain is also retained, but it is not part of the compact logical core:

$$\Omega' \xrightarrow{\mathcal{H}_{acc}=0,\ H_{diag}=0} 9 \times 3 \xrightarrow{a \neq b,\ d \text{ idempotent}} 6 \times 1 \xrightarrow{\min \mathrm{Assoc}_{000}} 2 \times 1 \xrightarrow{\text{pure } C/J} 1.$$

It displays the PAB/row-complement ambiguity before the undrifted C/J separation; Theorem 12.3 gives the shorter selection proof in which pure C/J, after the preceding three selectors, already forces PAB.

0.3 Landscape atlases

The associativity statistic

$$\mathrm{Assoc}(x) = \#\{(u, v, w) : (u \cdot v) \cdot w = u \cdot (v \cdot w)\}$$

has global range

$$63 \leq \mathrm{Assoc} \leq 597.$$

Its exact partition function has 167 nonzero coefficients, mean 245, variance 27820/27, and mode 237 with coefficient $426,317,976$. PAB has

$$\mathrm{Assoc}(\mathrm{PAB}) = 219.$$

The hidden-continuation atlas uses local signed contributions h_q, their total

$$H_{tot} = 3 \sum_{q \in S^5} h_q,$$

and the negative local count N_-. The global theorem gives

$$\min_{\Omega'} H_{tot} = -2268, \qquad \max_{\Omega'} H_{tot} = 7302, \qquad \max_{N_-=0} H_{tot} = 7020.$$

PAB and row-complement lie on this pure frontier; the twelve points above it are impure signed-cancellation maximizers.

0.4 Linearization after selection

After selection, PAB has a compact operator theory. Every left translation has image a row fiber and rank 3. The positive composition envelope has 51 maps, decomposed into three row-target components of size 17, and its linear span has dimension 27. The endomorphism monoid consists of the six simultaneous symbol permutations and three diagonal collapses. The binary term algebra has normal form

$$T_2(\mathrm{PAB}) \cong \{x, y\} \times \mathcal{S}_{21} \times \mathcal{U}_{15}, \qquad |T_2(\mathrm{PAB})| = 630.$$

The vector-space carrier identifies with matrix units,

$$k[M] \cong \mathrm{Mat}_3(k), \qquad e_{(r,c)} \leftrightarrow E_{rc}.$$

This carrier supports the row-contraction bridge

$$\mu_{cross}(A, B) = AJ_3B^\top = (A\mathbf{1})(B\mathbf{1})^\top,$$

the compact carrier-level type $SU(3) \times SU(2) \times U(1)$, and a signed symplectic lift of the pure C/J directed-edge geometry.

0.5 Proof architecture and support

The proof has three support modes.

result family	role	support
normal form, access-zero, diagonal compression, compact PAB selector	hand finite theorems	text; L2 tables audit survivor counts
associativity range, endpoints, partition function	finite landscape atlas	compact `tables/` plus L1 verifier
global hidden-continuation frontier	global frontier certificate	separate `mathcal_H/` bundle
operator, endomorphism, composition, T_2 structure	post-selection structure	text plus L3 realization/transition/audit tables
carrier Lie and signed symplectic structures	downstream bridge facts	bridge audit tables

The distributed support corpus keeps these roles separate:

`scripts/`, `tables/`, `mathcal_H/`.

The compact manuscript-facing check is

```
python3 -S scripts/verify_preprint_support.py
```

The global hidden-continuation bundle is checked separately from inside `mathcal_H/`.

0.6 Positioning

The first selection layer uses standard finite information observables in the language of Landauer, Bennett, Shannon, and Cover–Thomas [19, 3, 4, 26, 10]. The finite magma, term-function, and clone-theoretic context is standard universal algebra [5, 8, 7, 24, 25, 28, 20]. The same-row rule is the three-point Steiner multiplication, in the classical nonassociative and quasigroup background [6, 11, 23, 9]. The certificate layer belongs to the tradition of computer-assisted finite mathematics [1, 2, 13, 21, 15, 16]. The carrier-level matrix, Lie, and symplectic terminology follows standard conventions [18, 12, 17, 14, 22].

0.7 Notation

notation	meaning
S	the field $\mathbb{F}_3$
M	the carrier $S \times S$
F_r	the row fiber $\{r\} \times S$
$M^\times$	the six off-diagonal points $\{(r,c) : r \neq c\}$
Ω'	the AX0+AX2, σ-equivariant landscape
(A, B, d)	normal coordinates in $S^{18} \times S^3$
$h_{a,b}$	column-blind cross-row rule with anchors $h(0,1) = a$, $h(0,2) = b$
Assoc	total number of associative triples in M^3
Assoc_{000}	local associativity statistic after fixing $d = 000$

notation	meaning
$\mathcal{H}_{acc}$	hidden-column access observable
H_{diag}	entropy of the diagonal output multiset
h_q, H_{tot}, N_-	hidden-continuation local contribution, total value, and negative local count
C, J	continuation and reversal maps on $M^\times$
$T_2(\mathrm{PAB})$	binary term-function algebra of PAB

Part I

The finite landscape

1 Carrier, axioms, and normal form

1.1 The row-graded carrier

Let

$$S = \mathbb{F}_3, \qquad M = S \times S.$$

The first coordinate is called the row and the second coordinate the column. The row fibers are $F_r = \{r\} \times S$.

The left-row output axiom says that the product of (r_1, c_1) and (r_2, c_2) always lies in F_{r_1}. Hence a multiplication in the family is determined by its output-column function.

The same-row off-diagonal axiom fixes the product inside a row when the two columns are distinct:

$$(r, i) \cdot (r, j) = (r, \overline{ij}) \qquad (i \neq j).$$

Here $\overline{ij}$ denotes the unique element of $S \setminus \{i, j\}$. This is the Steiner multiplication on a three-point row fiber [23, 9].

1.2 Normal coordinates

The remaining freedom has two kinds. First, cross-row products are free in their output column. By σ-equivariance, it is enough to record two representative cross-row tables, denoted A and B, each with nine entries. Second, same-point products in a row are governed by a diagonal vector

$$d = (d_0, d_1, d_2) \in S^3.$$

The normal coordinates are therefore

$$(A, B, d) \in S^9 \times S^9 \times S^3.$$

1.3 Normal form theorem

Theorem 1.1. The active AX0+AX2, σ-equivariant landscape has normal form

$$\Omega' \cong S^{18} \times S^3 \cong S^{21}.$$

Consequently

$$|\Omega'| = 3^{21} = 10\,460\,353\,203.$$

Proof. AX0 fixes the output row of every product. AX2 fixes all same-row off-diagonal output columns. For cross-row products, σ-equivariance reduces the row-pair data to the two ordered representatives $(0,1)$ and $(0,2)$. Each representative has 3×3 column-input entries, giving 18 cross-row coordinates. The same-point diagonal sector contributes 3 coordinates. These coordinates are independent, so the landscape is S^{21}. □

2 Effective symmetry and orbit count

The resulting family has a residual effective symbol-relabelling symmetry. Because the landscape is already written in σ-equivariant normal coordinates, the effective action on these coordinates is generated by the remaining nontrivial relabelling class. We denote this residual action by

$$G_{\text{eff}} \cong C_2.$$

The diagonal sector still carries the ordinary symbol-relabelling compression

$$|\Delta/\operatorname{Sym}(S)| = 15.$$

Proposition 2.1. The effective orbit count of the full normal-coordinate landscape is

$$|\Omega'/G_{\text{eff}}| = \frac{3^{21} + 3^{10}}{2}.$$

Proof. The computation is Burnside's lemma for the residual effective action on normal coordinates. The identity contributes 3^{21}. The nontrivial effective symmetry has fixed locus of size 3^{10}. Dividing by $|G_{\text{eff}}| = 2$ gives the formula. □

The orbit count is a landscape compression.

3 Hidden-column access

The first selection functional is information-theoretic. It measures whether the cross-row rule has functional dependence on hidden column coordinates after the row pair is known. Its vanishing is the Landauer-facing condition that the cross-row channel admits a realization without querying column input.

3.1 Column-blind cross-row rules

A cross-row output rule is **column-blind** if its output column depends only on the two row coordinates:

$$g(r_1, c_1, r_2, c_2) = h(r_1, r_2).$$

Equivalently, after the ordered row pair is fixed, the output column is constant over the nine column inputs.

By σ-equivariance, a column-blind rule is determined by two values

$$a = h(0,1), \qquad b = h(0,2).$$

We call (a,b) the cross-row anchors and write $h_{a,b}$ for the corresponding rule. The remaining ordered row pairs are forced by σ-shift:

$$h(r, r+1) = r + a, \qquad h(r, r+2) = r + b$$

with arithmetic in $\mathbb{F}_3$. There are $3^2 = 9$ column-blind cross-row rules.

3.2 Hidden-column access as a Landauer-facing observable

Let

$$\Sigma_{cross} = \{(r_1, c_1, r_2, c_2) \in S^4 : r_1 \neq r_2\}$$

with the uniform distribution, and let

$$C_{out} = g(R_1, C_1, R_2, C_2)$$

be the output-column random variable. Define the hidden-column access observable by

$$\boxed{\mathcal{H}_{acc}(g) = I(C_{out}; C_1, C_2 \mid R_1, R_2).}$$

Logarithms are base 2, so the value is measured in bits.

This is a structural information observable. It measures how much hidden-column information is functionally retained in the cross-row output once the row pair is fixed. The Landauer reading is that hidden column information which is read and later discarded by an irreversible implementation carries thermodynamic cost [19, 3, 4]. Therefore $\mathcal{H}_{acc} = 0$ selects exactly those cross-row rules for which that cost is structurally avoidable: the output can be computed without consulting the hidden columns.

The zero condition is used in this implementation-independent structural sense: the column-read burden can be avoided altogether.

3.3 Access-zero theorem

Theorem 3.1. For a cross-row rule g,

$$\mathcal{H}_{acc}(g) = 0 \quad \Longleftrightarrow \quad g(r_1, c_1, r_2, c_2) = h(r_1, r_2).$$

There are exactly nine column-blind cross-row rules.

Proof. Conditional mutual information vanishes exactly when

$$C_{out} \perp (C_1, C_2) \mid (R_1, R_2).$$

Since C_{out} is a deterministic function of (R_1, C_1, R_2, C_2), conditional independence is equivalent to constancy in (C_1, C_2) for every fixed ordered row pair (R_1, R_2). That is precisely column-blindness.

The two representative cross-row row pairs $(0, 1)$ and $(0, 2)$ each receive one output-column value in S, giving $3^2 = 9$ σ-equivariant column-blind rules. □

This theorem is a landscape result and also the first reduction in the selection chain.

4 Associativity master formula

4.1 The associativity statistic

For $x \in \Omega'$, define

$$\mathrm{Assoc}(x) = \#\{(u, v, w) \in M^3 : (u \cdot v) \cdot w = u \cdot (v \cdot w)\}.$$

Since $|M| = 9$, this statistic lies between 0 and 729.

Theorem 4.1. In normal coordinates,

$$\mathrm{Assoc} = 3\, rawAssoc,$$

where $rawAssoc$ is a normalized sum over $3^5 = 243$ triples and decomposes as

$$rawAssoc = rawRRR + rawRRS + rawRSR + rawSRR + rawDIST.$$

The five blocks correspond to the row patterns

$$RRR, \qquad RRS, \qquad RSR, \qquad SRR, \qquad DIST.$$

Proof. By σ-equivariance the first row coordinate of a triple may be normalized to 0. The two remaining row variables and the three column variables give 3^5 normalized cases; the factor 3 restores the three translates. The displayed blocks are the five equality patterns of the three rows. □

4.2 The same-row block

The all-same-row block depends only on the diagonal vector d. Let

$$E(d) = \#\{(i,j) : i \neq j,\ d_i = d_j\}, \qquad N(d) = \#\{(i,j) : i \neq j,\ d_i = j,\ d_j = j\}.$$

Lemma 4.2. The same-row contribution is

$$rawRRR(d) = 9 + E(d) + 2N(d).$$

The four diagonal classes give raw same-row values

$$9, \qquad 11, \qquad 13, \qquad 19.$$

In particular,

$$rawRRR(000) = 19.$$

Proof. In one row, unequal column pairs use the Steiner rule and equal pairs use d. The 27 triples split into a baseline family plus the two diagonal-sensitive families counted by $E(d)$ and $N(d)$. This gives the formula and the four class values. □

4.3 Column-blind profile at $d = 000$

Let $h_{a,b}$ be the column-blind rule

$$h(r, r+1) = r + a, \qquad h(r, r+2) = r + b.$$

At $d = 000$, same-point products satisfy $(r,c)^2 = (r,r)$.

Proposition 4.3. On the column-blind, $d = 000$ stratum,

$$\begin{aligned}(rawRRR, rawRRS, rawRSR, rawSRR, rawDIST)(a,b,000)\\ = \big(19,\ 9[a=0] + 9[b=0],\ 9[a=0] + 9[b=0],\\ 54,\ 54[a=b]\big).\end{aligned}$$

Consequently

$$\boxed{\mathrm{Assoc}_{000}(h_{a,b}) = 3\,(73 + 18[a=0] + 18[b=0] + 54[a=b])\,.}$$

Proof. The RRR value is Lemma 4.2. In the RRS and RSR patterns, the same-row step must return the cross-row anchor; at $d = 000$ this contributes 9 exactly when the relevant anchor is 0. In SRR, the final cross-row output ignores same-row column processing, giving all 54 normalized triples. In $DIST$, the two association orders compare the two cross-row anchors, so all 54 triples survive exactly when $a = b$. □

The nine column-blind rules split as follows:

class	representatives	profile	Assoc_{000}
anchored trivial	$(0,0)$	$(19,18,18,54,54)$	489
off-anchor trivial	$(1,1),(2,2)$	$(19,0,0,54,54)$	381
partly anchored nondegenerate	$(0,1),(0,2),(1,0),(2,0)$	$(19,9,9,54,0)$	273
fully off-anchor nondegenerate	$(1,2),(2,1)$	$(19,0,0,54,0)$	219

The fourth class consists of PAB and row-complement.

5 Global associativity range

5.1 Exact range

Lemma 5.1 (Associativity atlas certificate protocol)**.** The global associativity atlas is used through five finite obligations.

1. **A1, block reduction.** The associativity statistic is reduced by σ-normalization to the five row-pattern blocks

$$rawAssoc = rawRRR + rawRRS + rawRSR + rawSRR + rawDIST$$

   of Theorem 4.1.
2. **A2, diagonal decomposition.** The same-row block is evaluated from the diagonal vector by Lemma 4.2, giving the four possible values

$$9, \quad 11, \quad 13, \quad 19.$$

3. **A3, fixed-diagonal cross-range certificates.** For each diagonal class, the certificate verifies the exact range of

$$rawCross = rawRRS + rawRSR + rawSRR + rawDIST.$$

4. **A4, endpoint witnesses and endpoint counts.** The endpoint certificates verify the normal-coordinate witnesses, endpoint counts, and orbit representatives for $\mathrm{Assoc} = 63$ and $\mathrm{Assoc} = 597$.
5. **A5, distribution consistency.** The partition-function certificate gives the full coefficient table for

$$Z(q) = \sum_{x \in \Omega'} q^{\mathrm{Assoc}(x)}$$

   and checks total mass, range, moments, and mode.

Any certificate satisfying A1–A5 proves the global associativity range, endpoint loci, and partition-function claims of Sections 5–6. This is a certified atlas statement for the ambient landscape; the compact PAB uniqueness proof in Theorem 12.3 is the separate hand spine in Part II. □

Theorem 5.2 (Global associativity atlas)**.** On Ω',

$$21 \leq rawAssoc \leq 199.$$

Equivalently,

$$63 \leq \mathrm{Assoc} \leq 597.$$

The finite certificate is organized by diagonal class:

diagonal class	rawRRR	certified rawCross range	rawAssoc consequence
permutation	9	12..180	21..189
two-valued, no fixed point	11	10..162	21..173
two-valued, with fixed point	13	12..174	25..187
constant	19	12..180	31..199

Here

$$rawCross = rawRRS + rawRSR + rawSRR + rawDIST.$$

The endpoint values are therefore $rawAssoc = 21$ and $rawAssoc = 199$, or $\mathrm{Assoc} = 63$ and $\mathrm{Assoc} = 597$.

Proof. The hand-readable reduction is Theorem 4.1 plus Lemma 4.2. After that reduction, the global claim is exactly the A3 cross-range obligation of Lemma 5.1. Adding the certified cross-row ranges to the four possible same-row values gives the displayed global range.

The endpoint witnesses, endpoint counts, orbit representatives, and partition-function consistency are the A4–A5 obligations of Lemma 5.1. They are recorded in `l1_assoc_distribution.csv`, `l1_extremal_loci_counts.csv`, `l1_extremal_loci_orbits.csv`, `l1_extremal_loci_solutions.csv`, and `l1_assoc_proof_targets.csv`, and checked by the compact L1 verifier. □

5.2 Global and local associativity roles

PAB has

$$\mathrm{Assoc}(\mathrm{PAB}) = 219,$$

while the landscape minimum is 63. The global statistic maps the ambient landscape. The selection argument later uses the conditional local statistic Assoc_{000}, after information compression and anchoring, to expose the final two-candidate ambiguity.

6 Endpoint loci and partition function

6.1 Endpoint loci

Theorem 6.1. The associativity endpoint loci have sizes

$$|\mathcal{M}_{63}| = 24, \qquad |\mathcal{M}_{63}/\mathrm{Sym}(S)| = 12,$$

and

$$|\mathcal{M}_{597}| = 6, \qquad |\mathcal{M}_{597}/\operatorname{Sym}(S)| = 3.$$

The minimum endpoint block profiles are

$$(11, 2, 2, 2, 4) \quad \text{and} \quad (9, 2, 2, 4, 4),$$

while the maximum endpoint block profile is

$$(19, 36, 36, 54, 54).$$

Proof. This is the A4 endpoint part of Lemma 5.1. The endpoint certificates record the exact solution counts by diagonal sector, orbit representatives, and explicit witnesses in normal coordinates. The minimum occurs in the permutation and two-valued/no-fixed diagonal classes. The maximum occurs in the constant diagonal class. The block profiles are obtained by evaluating the five normalized associativity blocks on the endpoint witnesses. □

6.2 Exact associativity partition function

Theorem 6.2. The exact polynomial

$$Z(q) = \sum_{x \in \Omega'} q^{\mathrm{Assoc}(x)}$$

has 167 nonzero coefficients and total mass

$$Z(1) = 3^{21}.$$

Its first two moments are

$$\mathbb{E}[\mathrm{Assoc}] = 245, \qquad \mathrm{Var}(\mathrm{Assoc}) = \frac{27820}{27}.$$

The mode occurs at

$$\mathrm{Assoc} = 237$$

with coefficient

$$426,317,976.$$

Proof. This is the A5 partition-function part of Lemma 5.1. The coefficient certificate `l1_assoc_distribution.csv` is the exact finite distribution of Assoc on Ω'. The total mass, range, mean, variance, and mode are computed directly from that table and checked by the compact L1 verifier. □

7 Hidden continuation

7.1 Local distance reduction

The hidden-continuation observable is a finite path-information defect. The continuation side keeps the two-step route data, while the left-translation side compares only the collapsed endpoint operators. Thus $h_q/2$ is the signed Hamming contrast between a path-resolved comparison and an endpoint-resolved comparison. Negative local values are precisely the local cancellation failures counted by N_-. The normalization $H_{tot} = 6rawH$ is the σ-orbit normalization used by the hidden-continuation certificate.

Its local index is

$$q = (b, t, a, e, f) \in S^5.$$

For a completed normal-form point $x \in \Omega'$, let M_s denote the local table read: M_1 and M_2 are the two cross-row tables, and M_0 is the same-row diagonal/complement rule. All row and column arithmetic below is in $S = \mathbb{F}_3$. Define

$$xy = M_b(a, e), \qquad yz = M_{t-b}(e - b, f - b), \qquad z_R = b + yz,$$

and

$$ep_L = M_t(xy, f), \qquad ep_R = M_b(a, z_R).$$

For $c \in S$, define the collapsed left-translation signature

$$L_c(i, j) = M_i(c, j) \qquad ((i, j) \in S^2).$$

For $s, \alpha, z \in S$, define the stepwise continuation signature

$$C_{s,\alpha,z}(i, j) = M_s(\alpha,\ s + M_{i-s}(z - s, j - s)) \qquad ((i, j) \in S^2).$$

The associated Hamming distances are

$$D_L(c, c') = \#\{(i, j) \in S^2 : L_c(i, j) \neq L_{c'}(i, j)\},$$

and

$$D_C(s, \alpha, z; s', \alpha', z') = \#\{(i, j) \in S^2 : C_{s,\alpha,z}(i, j) \neq C_{s',\alpha',z'}(i, j)\}.$$

Direct local definition. For each of the nine test pairs $(i, j) \in S^2$, set

$$\delta^C_{ij}(q; x) = \mathbf{1}\{C_{t,xy,f}(i, j) \neq C_{b,a,z_R}(i, j)\}\,,$$

and

$$\delta^L_{ij}(q; x) = \mathbf{1}\{L_{ep_L}(i, j) \neq L_{ep_R}(i, j)\}\,.$$

The nine direct local summands are

$$\Delta_{ij}(q; x) = 2\big(\delta^C_{ij}(q; x) - \delta^L_{ij}(q; x)\big),$$

displayed as

	$j = 0$	$j = 1$	$j = 2$
$i = 0$	Δ_{00}	Δ_{01}	Δ_{02}
$i = 1$	Δ_{10}	Δ_{11}	Δ_{12}
$i = 2$	Δ_{20}	Δ_{21}	Δ_{22}

and the direct hidden-continuation contribution is

$$h_q(x) = \sum_{(i,j)\in S^2} \Delta_{ij}(q; x).$$

Theorem 7.1 (Local distance reduction)**.** For every $x \in \Omega'$ and $q \in S^5$,

$$\boxed{h_q(x) = 2\,(D_C(t, xy, f; b, a, z_R) - D_L(ep_L, ep_R))\,.}$$

Consequently the local direct definition and the Hamming-distance formula define the same global observables $rawH$ and H_{tot}.

Proof. By definition, $D_C(t, xy, f; b, a, z_R)$ is the sum of the nine indicators $\delta^C_{ij}(q; x)$, and $D_L(ep_L, ep_R)$ is the sum of the nine indicators $\delta^L_{ij}(q; x)$. Subtracting these two Hamming distances and multiplying by 2 gives exactly the displayed sum of the nine direct local summands Δ_{ij}. The symbolic reduction is recorded by the certificate handles `S6-RED` and `S6-CERT-IF`. □

Then

$$rawH(x) = \sum_{q \in S^5} \frac{h_q(x)}{2}, \qquad H_{tot}(x) = 6\, rawH(x) = 3 \sum_{q \in S^5} h_q(x),$$

and

$$\boxed{N_-(x) = 3\,\#\{q \in S^5 : h_q(x) < 0\}.}$$

Thus purity is the finite system

$$N_-(x) = 0 \quad \Longleftrightarrow \quad h_q(x) \geq 0 \quad (q \in S^5).$$

7.2 Global frontier theorem

The global hidden-continuation theorem is supported by four finite certificate obligations.

handle	obligation	result
H1	unrestricted full-landscape range	$rawH \in [-378, 1217]$, endpoint counts $8, 12$
H2	pure-frontier exclusion	$(h_q \geq 0\ \forall q) \wedge rawH \geq 1171$ is UNSAT
H3	pure-frontier witnesses	PAB and row-complement have $rawH = 1170$, $N_- = 0$
H4	signed-cancellation spike	exactly twelve points above the pure frontier, all with $(rawH, N_-) = (1217, 3)$

Theorem 7.2 (Global frontier of $\mathcal{H}$)**.** On $\Omega' = S^{21}$:

1. The unrestricted range is

$$\boxed{\min H_{tot} = -2268, \qquad \max H_{tot} = 7302.}$$

   Equivalently, $rawH \in [-378, 1217]$. The endpoint multiplicities are 8 and 12.

2. The pure frontier is

$$\boxed{\max\{H_{tot} : N_- = 0\} = 7020.}$$

3. The region above the pure frontier is exactly the unrestricted maximum layer:

$$\boxed{\{H_{tot} > 7020\} = \{H_{tot} = 7302\}.}$$

   This layer has 12 points, all with

$$rawH = 1217, \qquad H_{tot} = 7302, \qquad N_- = 3,$$

$$H_{pos} = 7308, \qquad H_{\text{neg_abs}} = 6, \qquad h_{\text{loc,min}} = -2, \qquad h_{\text{loc,max}} = 18.$$

Proof. The analytic reduction is Theorem 7.1. H1 covers all 3^{21} normal-form words and gives the unrestricted range. For the pure frontier, $N_- = 0$ is the system $h_q \geq 0$ for all $q \in S^5$, and $H_{tot} > 7020$ is equivalent to $rawH \geq 1171$. H2 proves

$$(h_q \geq 0 \ \forall q \in S^5) \wedge rawH \geq 1171$$

UNSAT on all 18540 depth-9 live nodes:

$$18540/18540 \text{ UNSAT}, \qquad SAT = 0, \qquad UNKNOWN = 0.$$

H3 supplies pure witnesses at $rawH = 1170$, and H4 classifies the full layer above the pure threshold as the twelve impure maximizers. □

7.3 PAB and row-complement on the pure frontier

Corollary 7.3. PAB and row-complement satisfy

$$H_{tot} = 7020, \qquad rawH = 1170, \qquad N_- = 0.$$

Thus both lie on the global pure hidden-continuation frontier.

Proof. Direct evaluation of

$$\text{PAB} = 11111111222222222000, \qquad \text{comp} = 22222222211111111000$$

gives the displayed values, and Theorem 7.2 proves that no pure point has larger H_{tot}. □

Part I has now supplied the ambient finite map: normal form, global associativity, and the hidden-continuation frontier. Part II turns from atlas data to the intrinsic selector. Its proof uses the access-zero and diagonal-compression reductions, then the directed-edge absorption geometry; the frontier theorem supplies the global placement of the two finalists.

Part II

Intrinsic finite selection

8 Selection core

The compact uniqueness core is

$$\boxed{\mathcal{H}_{acc} = 0, \qquad H_{diag} = 0, \qquad \text{diagonal idempotence}, \qquad \text{pure } C/J.}$$

The order of the four conditions mirrors the row/column geometry of Ω'. Zero hidden-column access makes cross-row output column-blind. Zero diagonal entropy makes the same-point diagonal sector constant. Diagonal idempotence fixes that constant to $d = 000$. Pure C/J then reads the absorption geometry on the six directed edges $M^\times$ and chooses the undrifted continuation/reversal pair.

The longer explanatory chain is

$$\Omega' \xrightarrow{\mathcal{H}_{acc}=0,\ H_{diag}=0} 9 \times 3 \xrightarrow{a \neq b,\ d \text{ idempotent}} 6 \times 1 \xrightarrow{\min \text{Assoc}_{000}} 2 \times 1 \xrightarrow{\text{pure } C/J} 1.$$

It records the local path-dependence diagnostic that exposes the two finalists, PAB and row-complement, before the pure C/J separation.

9 Diagonal compression

The access-zero theorem has already compressed the cross-row sector to the nine column-blind rules $h_{a,b}$. The remaining information-compression step acts on the same-point diagonal data.

For a diagonal triple

$$d = (d_0, d_1, d_2) \in S^3,$$

let

$$p_s = \frac{1}{3}\#\{i : d_i = s\}.$$

Define the diagonal Shannon observable by

$$\boxed{H_{diag}(d) = -\sum_{s \in S} p_s \log_2 p_s.}$$

The possible levels are:

diagonal type	multiplicities	H_{diag}
constant	$(3, 0, 0)$	0
two-valued	$(2, 1, 0)$	$\log_2 3 - \frac{2}{3}$
permutation	$(1, 1, 1)$	$\log_2 3$

Theorem 9.1. For diagonal triples,

$$H_{diag}(d) = 0 \quad \Longleftrightarrow \quad d_0 = d_1 = d_2.$$

Thus diagonal compression reduces the diagonal sector from 27 choices to

$$000, \qquad 111, \qquad 222.$$

Proof. Entropy of a finite distribution vanishes exactly when the distribution is concentrated on one symbol. For the empirical distribution of (d_0, d_1, d_2), this is precisely constancy of the diagonal triple. □

Together with the access-zero theorem of Section 3, this reduces the selection domain to 9×3.

10 Anchors and local path-dependence

10.1 Nondegenerate anchors

Proposition 10.1. Among column-blind cross-row rules $h_{a,b}$, the condition

$$a \neq b$$

leaves exactly six rules.

Proof. There are nine pairs $(a, b) \in S^2$. The three diagonal pairs (a, a) do not distinguish the two cross-row representatives. Removing them leaves $3^2 - 3 = 6$ nondegenerate rules. □

Proposition 10.2. After diagonal compression, diagonal idempotence selects the unique constant diagonal

$$d = 000.$$

Proof. The compressed diagonal sector is $\{000, 111, 222\}$. Diagonal idempotence requires the product $(r, r) \cdot (r, r)$ to return (r, r), which fixes the constant diagonal 000 in the normalized coordinates. □

10.2 Local associativity at $d = 000$

Proposition 10.3. With $d = 000$ fixed, the six nontrivial column-blind rules have local associativity values

$$\mathrm{Assoc}_{000} = 219$$

for exactly two rules, and

$$\mathrm{Assoc}_{000} = 273$$

for the remaining four. The two local minima are PAB and row-complement.

The local-minimum profile is

$$(19, 0, 0, 54, 0),$$

whereas the four discarded rules have profile

$$(19, 9, 9, 54, 0).$$

The difference comes from the additional RRS and RSR coincidences in the discarded rules.

Proof. Proposition 4.3 gives

$$\mathrm{Assoc}_{000}(h_{a,b}) = 3\left(73 + 18[a = 0] + 18[b = 0] + 54[a = b]\right).$$

On the nondegenerate stratum $a \neq b$, this value is minimized exactly when neither anchor is 0, i.e. at $(a, b) = (1, 2)$ and $(2, 1)$. These are PAB and row-complement, and both have value 219. The four remaining nondegenerate pairs contain one zero anchor and have value 273. The certificate table `l2_path_dependence_table.csv` and verifier `verify_l2_selection.py` record the same six-row audit. □

This step is local and conditional. It is applied only after information compression, nondegenerate anchoring, and diagonal anchoring.

11 Directed-edge geometry

Let

$$M^{\times} = \{(r, c) \in M : r \neq c\}$$

be the six-point directed-edge set. Each $(r, c) \in M^{\times}$ is interpreted as the directed edge from r to c in the triangle on vertex set S.

Define

$$C(r, c) = (c, \overline{rc}), \qquad J(r, c) = (c, r).$$

The map C advances an oriented edge through the third vertex, and J reverses orientation.

Theorem 11.1. On $M^{\times}$,

$$C^3 = 1, \qquad J^2 = 1, \qquad JCJ = C^{-1}.$$

Thus C and J generate the dihedral geometry of the directed triangle.

Proof. Applying C three times traverses

$$(r, c) \mapsto (c, \overline{rc}) \mapsto (\overline{rc}, r) \mapsto (r, c).$$

Applying J twice reverses orientation twice. Finally,

$$JCJ(r, c) = JC(c, r) = J(r, \overline{rc}) = (\overline{rc}, r) = C^{-1}(r, c).$$

The table `l3_CJ_finite_relations.csv` audits the finite relation checks. □

12 Pure C/J and the selection theorem

12.1 Absorption and absorption transitions

For $x \in M^\times$, define the absorption set

$$\mathrm{Abs}(x) = \{y \in M^\times \setminus \{x\} : x \cdot y = x\}.$$

For a column-blind rule $h_{a,b}$ at $d = 000$, same-row right arguments never absorb an off-diagonal x. Thus absorption is governed by the cross-row equation

$$h(r_x, r_y) = c_x.$$

A candidate has a two-transition absorption geometry if every $x \in M^\times$ has exactly two absorbed neighbours. In that case the two absorbed-neighbour maps define AbsTrans. The pure C/J criterion requires

$$\mathrm{AbsTrans} = \{C, J\}$$

as literal maps on $M^\times$, with no $C^{\pm 1}$-drift.

12.2 The final two candidates

After local path-dependence, the two remaining rules are PAB and row-complement.

Lemma 12.1. For PAB,

$$\mathrm{AbsTrans}_{PAB} = \{C, J\}.$$

For row-complement,

$$\mathrm{AbsTrans}_{comp} = \{C^{-1}, C^{-1}J\}.$$

Proof. Let $x = (r, c) \in M^\times$.

For PAB, $h(r_x, r_y) = r_y$. The absorption equation is

$$r_y = c.$$

Hence the two absorbed neighbours of (r, c) are the two off-diagonal points in row c:

$$(c, r), \qquad (c, \overline{rc}).$$

These are precisely

$$J(r, c) = (c, r), \qquad C(r, c) = (c, \overline{rc}).$$

So $\mathrm{AbsTrans}_{PAB} = \{C, J\}$.

For row-complement, $h(r_x, r_y) = \overline{r_x r_y}$. The absorption equation is

$$\overline{r r_y} = c,$$

so

$$r_y = \overline{rc}.$$

The two absorbed neighbours are therefore

$$(\overline{rc}, r), \qquad (\overline{rc}, c),$$

which are

$$C^{-1}(r, c) = (\overline{rc}, r), \qquad C^{-1}J(r, c) = (\overline{rc}, c).$$

Thus $\mathrm{AbsTrans}_{comp} = \{C^{-1}, C^{-1}J\}$. □

12.3 Final selector among the two local finalists

Theorem 12.2. Among the final two candidates, the pure C/J directed-edge criterion selects PAB.

Proof. By Lemma 12.1, PAB realizes the undrifted pair $\{C, J\}$. Row-complement realizes $\{C^{-1}, C^{-1}J\}$, whose continuation and reversal are drifted by C^{-1}. The pure criterion accepts only the literal undrifted pair. The audit table `l2_pure_CJ_survivors.csv` records the same comparison. □

12.4 Compact uniqueness core

Theorem 12.3. The four conditions

$$\mathcal{H}_{acc} = 0, \qquad H_{diag} = 0, \qquad \text{diagonal idempotence}, \qquad \text{pure } C/J$$

select PAB uniquely inside Ω'.

Proof. The first condition gives column-blindness by Theorem 3.1. Hence the cross-row rule has the form

$$h(r, r+1) = r + a, \qquad h(r, r+2) = r + b$$

for a unique pair $(a, b) \in S^2$. The second condition gives a constant diagonal by Theorem 9.1, and diagonal idempotence fixes this constant to $d = 000$ by Proposition 10.2.

It remains to impose pure C/J. Let $x = (r, c) \in M^\times$ and write $\delta = c - r \in \{1, 2\}$. Same-row right arguments do not absorb x at $d = 000$, so absorption is governed by the cross-row equation

$$x \cdot y = x \quad \Longleftrightarrow \quad h(r, r_y) = c.$$

For the two possible cross rows this becomes

$$r_y = r + 1 \Rightarrow a = \delta, \qquad r_y = r + 2 \Rightarrow b = \delta.$$

The pure C/J criterion requires the two absorbed neighbours of $x = (r, c)$ to be exactly

$$J(r, c) = (c, r), \qquad C(r, c) = (c, \overline{rc}).$$

Both lie in row c. Therefore, when $\delta = 1$, absorption must occur in row $r + 1 = c$, forcing $a = 1$. When $\delta = 2$, absorption must occur in row $r + 2 = c$, forcing $b = 2$. Thus

$$(a, b) = (1, 2).$$

This is precisely the PAB cross-row rule, and $d = 000$ has already been fixed. Hence the four conditions select PAB uniquely. □

The audit table `l2_minimal_selector_sets.csv` records the same compact core as a finite audit of the survivor condition.

12.5 Full explanatory selection chain

Proposition 12.4. The finite explanatory chain

$$\Omega' \xrightarrow{\mathcal{H}_{acc}=0,\ H_{diag}=0} 9\times 3 \xrightarrow{a\neq b,\ d \text{ idempotent}} 6\times 1 \xrightarrow{\min \mathrm{Assoc}_{000}} 2\times 1 \xrightarrow{\text{pure } C/J} 1$$

has unique survivor PAB.

Proof. The first arrow is Theorems 3.1 and 9.1. The second arrow is Propositions 10.1 and 10.2. The third arrow is Proposition 10.3. The final arrow is Theorem 12.2. The audit table `l2_selection_chain.csv` records the survivor count at each stage, and the compact L2 verifier checks the displayed chain. □

This chain is logically subsumed by Theorem 12.3. After column-blindness, diagonal compression, and diagonal idempotence, the pure C/J condition alone forces $(a, b) = (1, 2)$. The $\min \mathrm{Assoc}_{000}$ arrow is retained only as a didactic intermediate: it exposes the PAB/row-complement ambiguity before the directed-edge orientation criterion separates them.

Part III
Linearization after selection

13 The selected PAB operation

13.1 PAB multiplication and ground fields

The selected multiplication is

$$(r_1, c_1)\cdot(r_2, c_2) = \begin{cases} (r_1, r_2), & r_1 \neq r_2, \\ (r_1, \overline{c_1 c_2}), & r_1 = r_2,\ c_1 \neq c_2, \\ (r_1, r_1), & r_1 = r_2,\ c_1 = c_2. \end{cases}$$

The three cases are cross-row multiplication, same-row off-diagonal Steiner multiplication, and diagonal anchoring.

The finite magma statements are set-theoretic. When the carrier is linearized, let k be a ground field and let $k[M]$ be the vector space with basis e_x, $x \in M$. Rank and image statements for individual finite maps are valid over any ground field. The finite composition monoid and term-function cardinalities are field-independent; linear span dimensions are dimensions over the chosen k. The left-translation span in Theorem 14.2 is characteristic-independent by the displayed normal-form proof. Spectral, Jordan, and fiber-algebra statements involving the factors $t-1$, $t+1$, or the idempotents with coefficients $1/2$ are stated under

$$\operatorname{char} k \neq 2.$$

The compact Lie and symplectic bridge statements are over $\mathbb{R}$ unless explicitly stated otherwise.

The carrier vector space admits the matrix-unit identification

$$k[M] \cong \mathrm{Mat}_3(k), \qquad e_{(r,c)} \leftrightarrow E_{rc}.$$

This identification is used for linear and bridge structures. PAB multiplication remains the finite multiplication displayed above.

13.2 Row-owner principle

For a nonempty term $t(x_1, \ldots, x_n)$, define its **owner** $\mathrm{ow}(t)$ to be the leftmost variable appearing in the term tree.

Lemma 13.1 (Row-owner lemma)**.** For every PAB term $t(x_1, \ldots, x_n)$,

$$row\big(t(x_1, \ldots, x_n)\big) = row\big(\mathrm{ow}(t)\big).$$

Proof. The assertion is immediate for a variable. If $t = u \cdot v$, then the left-row output axiom gives

$$row(t) = row(u \cdot v) = row(u).$$

By induction, $row(u)$ is the row of the leftmost variable of u, which is also the leftmost variable of $u \cdot v$. □

This lemma is the structural source of the owner coordinate in the binary term algebra of Section 18 and of the row-target law for compositions of left translations.

14 Left regular operators

For $x \in M$, define the left translation

$$L_x(y) = x \cdot y.$$

14.1 Row-fiber normal form

Proposition 14.1. For $x = (r, c)$, the left translation has the explicit form

$$L_{(r,c)}(s, d) = \begin{cases} (r, s), & s \neq r, \\ (r, r), & s = r,\ d = c, \\ (r, \overline{cd}), & s = r,\ d \neq c. \end{cases}$$

In particular,

$$L_{(r,c)}(M) = F_r.$$

Proof. If $s \neq r$, the PAB cross-row rule gives $(r, c) \cdot (s, d) = (r, s)$. If $s = r$ and $d = c$, diagonal anchoring gives (r, r). If $s = r$ and $d \neq c$, the same-row Steiner rule gives $(r, \overline{cd})$. These three cases also show that the image contains all three points of F_r. □

14.2 Rank and span

Theorem 14.2. For every $x \in M$,

$$\operatorname{rank} L_x = 3, \qquad \operatorname{Im} L_x = F_{row(x)}.$$

Moreover,

$$\dim \operatorname{span}\{L_x : x \in M\} = 9.$$

Proof. Proposition 14.1 shows that each L_x maps onto the three-point row fiber $F_{row(x)}$. After linearization, the three basis vectors of that row fiber occur among the columns of the matrix of L_x, so the rank is 3.

It remains to prove the span dimension. Operators with different target rows have disjoint codomain support, so their spans are independent. Fix a row r. The three operators $L_{(r,c)}$, $c \in S$, are independent because their restrictions to kF_r are, after a relabelling of S, the three matrices

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

and these are visibly linearly independent. Thus each row contributes dimension 3, and the total span dimension is $3 \cdot 3 = 9$. The table `l3_operator_spectrum_rank.csv` audits the nine-row calculation. □

14.3 Spectral dichotomy

Assume $\operatorname{char} k \neq 2$.

Theorem 14.3. If $x = (r, r)$ is diagonal, then

$$\mu_{L_x}(t) = t(t-1)(t+1),$$

and L_x is semisimple. If $x = (r, c)$ with $r \neq c$, then

$$\mu_{L_x}(t) = t^2(t-1)(t+1),$$

with zero-primary Jordan structure

$$J_2(0) \oplus J_1(0)^{\oplus 5}.$$

Proof. Write $V = k[M]$ and $V_r = kF_r$. Every $L_{(r,c)}$ maps V onto V_r. Hence all nonzero spectral data are controlled by the restriction

$$T_{r,c} = L_{(r,c)}|_{V_r} : V_r \to V_r.$$

If $c = r$, the map $T_{r,r}$ fixes $e_{(r,r)}$ and swaps the two off-diagonal basis vectors in V_r. Thus its minimal polynomial is $(t-1)(t+1)$. Since $T_{r,r}$ is invertible, $V = \ker L_{(r,r)} \oplus V_r$, and $L_{(r,r)}$ has minimal polynomial $t(t-1)(t+1)$ and no nilpotent zero-primary extension.

If $c \neq r$, the restriction $T_{r,c}$ has eigenvalues $0, 1, -1$ on V_r. Its kernel is one-dimensional, so

$$\operatorname{Im} L_{(r,c)} \cap \ker L_{(r,c)} = \ker T_{r,c}$$

is one-dimensional. This produces one length-two Jordan block at eigenvalue 0, while the nonzero ± 1 part remains semisimple. Therefore the minimal polynomial is $t^2(t-1)(t+1)$. The table `l3_operator_spectrum_rank.csv` audits the spectral/Jordan calculation for all nine left translations. □

15 Composition envelope and directed-edge representation

15.1 Fixed-row normal forms for compositions

Let $\mathrm{Mon}^+\langle L_x\rangle$ be the positive composition monoid generated by the nine left translations. For $r \in S$, let $\mathcal{W}_r$ be the component of positive compositions whose output lies in F_r.

For fixed r, define

$$\tau_{r,c} : S \to S, \qquad \tau_{r,c}(d) = \begin{cases} r, & d = c, \\ \overline{cd}, & d \neq c. \end{cases}$$

This is the restriction of $L_{(r,c)}$ to F_r, after identifying F_r with its column set. Let

$$\mathcal{P}_r = \langle \tau_{r,c} : c \in S\rangle.$$

In normalized row 0,

$$\mathcal{P}_0 = \{012, 021, 200, 100, 011, 022\}.$$

For $\psi \in \mathcal{P}_r$ and $c \in S$, set

$$N_{r,\psi,c}(s,d) = \begin{cases} (r, \psi(\tau_{r,c}(d))), & s = r, \\ (r, \psi(s)), & s \neq r. \end{cases}$$

Also let $K_{r,t}$ be the constant map to (r,t).

Proposition 15.1 (Fixed-row composition normal form)**.** For every $r \in S$,

$$\boxed{\mathcal{W}_r = \{N_{r,\psi,c} : \psi \in \mathcal{P}_r,\ c \in S\} \cup \{K_{r,t} : t \in S\}.}$$

The first set contains 14 distinct nonconstant maps, the second contains 3 constants, and therefore

$$|\mathcal{W}_r| = 17.$$

Proof. A positive composition has a leftmost generator, and by the row-owner lemma its target row is the row r of that generator. If every generator in the word has target row r, the rightmost generator is some $L_{(r,c)}$; it applies $\tau_{r,c}$ on F_r and sends every other row F_s to column s. The remaining left prefix is an element $\psi \in \mathcal{P}_r$, yielding $N_{r,\psi,c}$. Conversely each such form is realized by a word for ψ followed by $L_{(r,c)}$.

If a generator to the right has target row $s \neq r$, the leftmost row-r generator sees a cross-row input and the word becomes constant; the three constants are realized by two-letter words and row-fiber prefixes. In row 0, the six-by-three list $\mathcal{P}_0 \times S$ has four twofold collisions and no other collisions, leaving $18 - 4 = 14$ nonconstant forms. Translation gives the same count for all rows. □

15.2 Composition envelope theorem

Theorem 15.2. The positive monoid generated by the left translations has size

$$|\,\mathrm{Mon}^+\langle L_x\rangle| = 51.$$

More precisely,

$$\mathrm{Mon}^+\langle L_x\rangle = \mathcal{W}_0 \sqcup \mathcal{W}_1 \sqcup \mathcal{W}_2, \qquad |\mathcal{W}_r| = 17.$$

Its linear envelope decomposes as

$$\mathcal{E} = W_0 \oplus W_1 \oplus W_2, \qquad \dim W_r = 9, \qquad \dim \mathcal{E} = 27,$$

and satisfies

$$W_r W_s \subseteq W_r.$$

Proof. Proposition 15.1 gives $|\mathcal{W}_r| = 17$, and the three target-row components are disjoint. Hence the positive composition monoid has $3 \cdot 17 = 51$ elements. If $A \in \mathcal{W}_r$ and $B \in \mathcal{W}_s$, then $A \circ B$ has target row r, proving $W_r W_s \subseteq W_r$.

For the span dimension, it suffices by symmetry to work in row 0. The certificate exposes nine normalized maps whose one-hot linearizations have a determinant -1 minor, and reduces the remaining eight maps to this basis by integer relations. Thus $\dim W_0 = 9$, and translating rows gives $\dim W_r = 9$ for all r. Disjoint codomain supports give the direct sum and $\dim \mathcal{E} = 27$. The tables `l3_composition_envelope.csv` and `l3_composition_product_law.csv` audit the normal forms, basis reductions, component counts, and row-target product law. □

15.3 The five-dimensional row-fiber algebra

Assume $\operatorname{char} k \neq 2$. For each row r, define

$$A_{5,r} = \langle L_{(r,c)}|_{kF_r} : c \in S \rangle \subseteq \operatorname{End}(kF_r).$$

Proposition 15.3. The algebra $A_{5,r}$ has dimension 5. For $r = 0$, with

$$B_1 = L_{(0,0)}|_{kF_0}, \qquad B_2 = L_{(0,1)}|_{kF_0}, \qquad B_3 = L_{(0,2)}|_{kF_0}, \qquad B_5 = B_2^2,$$

one has the basis

$$B_1, \quad B_2, \quad B_3, \quad I, \quad B_5.$$

Furthermore,

$$Z(A_{5,r}) = kI,$$

$A_{5,r}$ has a two-dimensional square-zero radical, and

$$A_{5,r}/\operatorname{Rad}(A_{5,r}) \cong k^3.$$

Proof. By σ-equivariance it is enough to check $r = 0$. The multiplication table of the five displayed basis elements closes and has dimension 5. The retained fiber-algebra certificate identifies the center as kI, the radical as a two-dimensional square-zero ideal, and the quotient as k^3. The supporting tables are `l3_fiber_algebra_5dim_multiplication.csv` and `l3_fiber_algebra_5dim_structure.csv`, checked by `verify_l3_fiber_algebra_5dim.py`. □

15.4 Directed-edge representation

Theorem 15.4. On the six directed edges,

$$\mathbb{R}[M^\times] \cong \mathbf{1} \oplus \operatorname{sign} \oplus 2\,\mathrm{std}.$$

For PAB the absorption matrix is

$$A_{abs} = (P_C + P_J)^\top$$

and has spectrum

$$\{2, 0, 0, 0, -1, -1\}.$$

Proof. The six directed edges carry the permutation representation of the directed triangle. Its character decomposition is $\mathbf{1} \oplus \mathrm{sign} \oplus 2\mathrm{std}$. PAB absorption is generated by the two pure transitions C and J, so the absorption matrix is $(P_C + P_J)^\top$. The tables `l3_absorption_transitions.csv` and `l3_absorption_spectrum.csv` audit the transition action and spectrum. □

16 Endomorphisms, intertwiners, and row contraction

16.1 Endomorphisms and intertwiners

For a map $\rho : S \to S$, write

$$\varphi_\rho(r, c) = (\rho(r), \rho(c))$$

when this formula defines a magma endomorphism. For $t \in S$, write

$$e_t(r, c) = (t, t)$$

for the collapse to the diagonal point (t, t).

For a row r, write the induced column operation on S as

$$c \star_r d = \begin{cases} r, & c = d, \\ \overline{cd}, & c \neq d. \end{cases}$$

Lemma 16.1 (row-fiber homomorphisms)**.** Let $\phi : S \to S$ satisfy $\phi(r) = t$ and

$$\phi(c \star_r d) = \phi(c) \star_t \phi(d) \qquad (c, d \in S).$$

Then either ϕ is the constant map to t, or ϕ is a permutation of S sending r to t.

Proof. By translating symbols, it is enough to take $r = t = 0$. Write

$$\phi(0) = 0, \qquad \phi(1) = u, \qquad \phi(2) = v.$$

The equations $0 \star_0 1 = 2$ and $0 \star_0 2 = 1$ give

$$v = 0 \star_0 u, \qquad u = 0 \star_0 v.$$

These two equations have exactly the three solutions

$$(u, v) = (0, 0), \qquad (u, v) = (1, 2), \qquad (u, v) = (2, 1).$$

They are respectively the constant map and the two permutations fixing 0, and each directly preserves $\star_0$. Translating back gives the claim. □

Theorem 16.2. The endomorphism monoid of the selected PAB magma is

$$\mathrm{End}(M, \cdot) = \{\varphi_\rho : \rho \in \mathrm{Sym}(S)\} \cup \{e_0, e_1, e_2\}.$$

Proof. The idempotents of PAB are exactly the diagonal points (r, r), since

$$(r, c)^2 = (r, r)$$

and this equals (r, c) only for $c = r$. Hence an endomorphism f sends diagonal points to diagonal points. Write

$$f(r, r) = (\rho(r), \rho(r)).$$

For $x = (r, c) \in F_r$, the identity $x^2 = (r, r)$ gives

$$f(x)^2 = f(r, r) = (\rho(r), \rho(r)),$$

so $f(x) \in F_{\rho(r)}$. Therefore

$$f(r, c) = (\rho(r), \phi_r(c))$$

for maps $\phi_r : S \to S$ satisfying $\phi_r(r) = \rho(r)$. Applying the endomorphism equation to pairs inside F_r shows that ϕ_r is a row-fiber homomorphism. By Lemma 16.1, each ϕ_r is either the constant map to $\rho(r)$ or a permutation of S sending r to $\rho(r)$.

Now impose cross-row multiplication. If $r \neq s$, then

$$f((r, c) \cdot (s, d)) = f(r, s) = (\rho(r), \phi_r(s)).$$

If $\rho(r) \neq \rho(s)$, the right side of the endomorphism equation is cross-row, so

$$f(r, c) \cdot f(s, d) = (\rho(r), \rho(s)),$$

and hence

$$\phi_r(s) = \rho(s).$$

If ρ is injective, this holds for every $r \neq s$, and together with $\phi_r(r) = \rho(r)$ gives $\phi_r = \rho$ for all r. Thus $\rho \in \mathrm{Sym}(S)$ and $f = \varphi_\rho$.

It remains to exclude nonconstant maps with noninjective ρ. Choose $r \neq s$ with $\rho(r) = \rho(s) = t$. The cross-row equation becomes

$$\phi_r(s) = \phi_r(c) \star_t \phi_s(d) \qquad (c, d \in S).$$

For fixed d, the map $a \mapsto a \star_t \phi_s(d)$ is not constant on S. Therefore ϕ_r cannot be a permutation, so Lemma 16.1 forces ϕ_r to be constant to t. Symmetrically, ϕ_s is constant to t.

Let u be the third row. If $\rho(u) \neq t$, comparing row u with the collapsed rows r and s gives

$$\phi_u(r) = t, \qquad \phi_u(s) = t.$$

This contradicts Lemma 16.1: ϕ_u is neither a permutation nor the constant map to $\rho(u)$. Hence $\rho(u) = t$, and then applying the preceding collapsed-pair argument to u and r forces ϕ_u to be constant to t. Thus $f = e_t$.

Conversely, every φ_ρ with $\rho \in \mathrm{Sym}(S)$ preserves the three cases in the PAB multiplication, and every collapse e_t is an endomorphism because $e_t(x \cdot y) = (t, t) = e_t(x) \cdot e_t(y)$. This gives exactly six permutation endomorphisms and three collapses. □

For an endomorphism φ, let Hom_φ denote the linear space of intertwiners satisfying

$$TL_x = L_{\varphi(x)}T \qquad (x \in M).$$

For automorphisms,

$$\mathrm{Hom}_{\varphi_\rho} = kP_\rho, \qquad \dim = 1.$$

For collapses,

$$\mathrm{Hom}_{e_r} = \{u\varepsilon : u \in \mathrm{span}\{d_r, o_r\}\}, \qquad \dim = 2.$$

Therefore

$$\sum_\varphi \dim \mathrm{Hom}_\varphi = 6 \cdot 1 + 3 \cdot 2 = 12.$$

The displayed intertwiner formulas are checked by substitution into the intertwining equations; the upper bounds and dimensions are audited in `l3_intertwiner_audit.csv`. Thus the table is an audit of the linear equations after the endomorphisms have been classified structurally.

16.2 Cross-row row contraction

Proposition 16.3. Under the carrier identification $k[M] \cong \mathrm{Mat}_3(k)$, the row-contraction extension of the cross-row PAB rule is

$$\mu_{cross}(A, B) = AJ_3B^\top = (A\mathbf{1})(B\mathbf{1})^\top.$$

On matrix units with cross rows,

$$E_{ij}J_3E_{kl}^\top = E_{ik}.$$

Proof. Since $J_3 = \mathbf{1}\mathbf{1}^\top$,

$$E_{ij}J_3E_{kl}^\top = (e_ie_j^\top)(\mathbf{1}\mathbf{1}^\top)(e_le_k^\top) = e_i(e_j^\top\mathbf{1})(\mathbf{1}^\top e_l)e_k^\top = E_{ik}.$$

For $i \neq k$, this is exactly the cross-row PAB rule $(i, j) \cdot (k, l) = (i, k)$. The table `l3_mat3_bridge_table.csv` audits the basis-level check and the coincidences with ordinary matrix multiplication. □

With the row-sum augmentation $\varepsilon(A) = A\mathbf{1}$,

$$\mu_{cross}(A, B) = \varepsilon(A)\varepsilon(B)^\top.$$

Thus the matrix bridge expresses the same column-blindness selected by $\mathcal{H}_{acc} = 0$: cross-row multiplication factors through row sums. If one extracts only the cross-row branch as a summand of the full bilinear PAB product on arbitrary matrices, the corresponding expression is

$$\pi_{off}\left(AJ_3B^\top\right).$$

The same-row PAB branch is the Steiner/diagonal rule on row fibers.

17 Hidden-continuation operator identity

The local hidden-continuation identity of Section 7 has an operator interpretation in the left-regular envelope. For a normalized triple indexed by q, the continuation distance D_C compares the two stepwise computations, while D_L compares the two collapsed left translations.

Theorem 17.1. For every local index $q \in S^5$,

$$\frac{h_q}{2} = D_C(t, xy, f; b, a, z_R) - D_L(ep_L, ep_R).$$

Equivalently, in the left-regular language this is the finite Hamming-distance contrast between

$$L_{xy}L_z \quad \text{and} \quad L_xL_{yz}$$

on the incremental side, and

$$L_{(xy)z} \quad \text{and} \quad L_{x(yz)}$$

on the collapsed side. All four operators lie in the same row-target component $W_{row(x)}$.

For PAB,

$$I_{tot} = 8904, \qquad B_{tot} = 1884, \qquad H_{tot} = 7020, \qquad N_- = 0.$$

The row-complement has the same pure hidden-continuation profile.

Proof. The first statement is Theorem 7.1 rewritten in operator language. Row-target compatibility follows from $W_r W_s \subseteq W_r$. The bridge audit `l3_H_bridge_pab_comp.csv` records the PAB and row-complement comparison; the finite check verifies that both have $H_{tot} = 7020$ and $N_- = 0$. The global frontier status follows from Theorem 7.2. □

The structural reading is one-sided path information. At the pure frontier, every local contribution is nonnegative: stepwise continuation never carries less finite distinction than the collapsed left-translation comparison. The only way to exceed the pure frontier is the twelve-point impure spike of Theorem 7.2, where signed local cancellation is present.

18 Binary term algebra

18.1 Binary sectors and normal signatures

A binary term function of PAB is a function $M^2 \to M$ realized by a term in variables x, y. The set of all such functions modulo equality is denoted $T_2(\mathrm{PAB})$.

By the row-owner lemma, every binary term has an owner variable, the leftmost variable in its term tree:

$$\mathrm{ow}(f) \in \{x, y\}.$$

Order the two inputs as

$$(o, n) = (\mathrm{owner}, \mathrm{nonowner}).$$

On the same-row sector, normalize the common row to 0 and define

$$s_f(i, j) = \mathrm{col}\big(f(o = (0, i), n = (0, j))\big), \qquad i, j \in S.$$

On the cross-row sector, normalize the owner row to 0 and the nonowner row to 1. The output column is independent of the nonowner column; define

$$u_f(i) = \mathrm{col}\big(f(o = (0, i), n = (1, j))\big), \qquad i \in S.$$

The opposite cross-row direction is determined by

$$u_f^-(i) = -u_f(-i).$$

Proposition 18.1 (Binary sector reduction)**.** Every binary term function has a normal signature

$$\Theta(f) = \big(\mathrm{ow}(f), s_f, u_f\big).$$

The same-row signature $s_f : S^2 \to S$ and cross unary signature $u_f : S \to S$ determine the term function on all row sectors.

Proof. The owner coordinate is Lemma 13.1. Variables have the displayed sector form. If $f = g \cdot h$, the owner of f is the owner of g. On same-row inputs, both subterms remain in input rows, so the output column is a function of the two normalized columns. On cross-row inputs, the right subterm is either in the nonowner row, giving a cross-row output independent of the nonowner column, or in the owner row, where the same-row PAB rule combines two unary owner-column signatures. The opposite cross-row direction follows by the PAB-compatible relabelling fixing the owner row and exchanging the two nonowner directions. □

18.2 Exposed finite factors

The realized same-row signatures form a 21-element set $\mathcal{S}_{21}$. The realized cross unary signatures form a 15-element set $\mathcal{U}_{15}$. The factorization index lists every element of both sets and gives realizing terms for the triples below. Thus the candidate binary normal forms are

$$\{x, y\} \times \mathcal{S}_{21} \times \mathcal{U}_{15}.$$

The finite transition tables record the induced pointwise PAB multiplication on the 21-state and 15-state factors.

18.3 Binary realization certificate

Lemma 18.2 (Binary term certificate protocol)**.** The binary term theorem is supported by four finite obligations.

obligation	content
T2-SIG	the factorization index exposes exactly $\mathcal{S}_{21}$ and $\mathcal{U}_{15}$
T2-REAL	every $(o, s, u) \in \{x, y\} \times \mathcal{S}_{21} \times \mathcal{U}_{15}$ has a realizing term
T2-TRANS	pointwise multiplication closes on the two factor transition systems
T2-MULT	the full 630×630 multiplication table is reconstructed from owner and factor transitions

The analytic part is the sector reduction and injectivity of Θ; the certificate supplies realization and closure of the finite factors.

18.4 Binary normal-form theorem

Theorem 18.3. The binary term-function algebra of PAB is finite and has size

$$|T_2(\mathrm{PAB})| = 630.$$

More precisely,

$$\boxed{T_2(\mathrm{PAB}) \cong \{x, y\} \times \mathcal{S}_{21} \times \mathcal{U}_{15}.}$$

Consequently

$$|T_2(\mathrm{PAB})| = 2 \cdot 21 \cdot 15 = 630.$$

The full binary multiplication table is recovered from the owner law and the two factor transition systems.

Proof. If $\Theta(f) = \Theta(g)$, then f and g agree on the same-row sector by equality of s, on the normalized cross-row sector by equality of u, and on the opposite cross-row sector by $u^-(i) = -u(-i)$. Hence Θ is injective.

Surjectivity is T2-REAL: the factorization index lists one realizing term for every triple in $\{x, y\} \times \mathcal{S}_{21} \times \mathcal{U}_{15}$. Therefore the number of binary term functions is $2 \cdot 21 \cdot 15 = 630$. The product law follows from $\mathrm{ow}(f \cdot g) = \mathrm{ow}(f)$ and the two finite factor transition systems; T2-MULT checks reconstruction of the full table. $\square$

The factorization separates owner, same-row, and cross-row data. It also marks the end of the finite term-function layer considered here: the all-arity term algebra is left for the open-problem list.

The same row/column coordinates used throughout the selector and term algebra now give a matrix-unit carrier. The next two sections record the continuous structures supported by that carrier and by the selected six-edge C/J geometry.

19 Carrier-level companion Lie package

This section records ordinary matrix and Lie structures on the vector-space carrier exposed after PAB selection [18, 12, 17]. Carrier-level means that the construction uses

$$k[M] \cong \mathrm{Mat}_3(k)$$

and the PAB-compatible simultaneous symbol-permutation action.

19.1 Carrier identification

The carrier vector space identifies with matrix units:

$$k[M] \cong \mathrm{Mat}_3(k), \qquad e_{(r,c)} \leftrightarrow E_{rc}.$$

PAB multiplication and ordinary matrix multiplication are two bilinear structures on this nine-dimensional carrier. Ordinary matrix multiplication gives

$$\mathfrak{gl}(3,k) = kI \oplus \mathfrak{sl}(3,k),$$

and the carrier decomposes as

$$k[M] = \mathbf{1} \oplus \mathbf{8}$$

under the adjoint matrix Lie package.

19.2 Compact and split $\mathrm{Sym}(S)$-stable branches

Over $\mathbb{R}$, the compact three-dimensional branch is

$$\mathfrak{so}(3) = \{X \in \mathfrak{sl}(3,\mathbb{R}) : X^\top = -X\}.$$

It is stable under simultaneous permutation conjugation

$$X \mapsto P_\rho X P_\rho^{-1}, \qquad \rho \in \mathrm{Sym}(S).$$

There is also a natural split comparator. With

$$\mathbb{R}^3 = \mathbf{1} \oplus V_{std}, \qquad V_{std} = \{v : v_0 + v_1 + v_2 = 0\},$$

the embedded algebra

$$\mathfrak{sl}(V_{std}) \subset \mathfrak{sl}(3,\mathbb{R})$$

is the subalgebra with zero row sums and zero column sums. It is $\mathrm{Sym}(S)$-stable and isomorphic to the split real $\mathfrak{sl}_2$. Compactness selects the $\mathfrak{so}(3)$ branch among these three-dimensional carrier-level branches.

19.3 Carrier-level compact matrix package

Theorem 19.1. In the real/complex compact-form setting, the carrier identification and the PAB-compatible $\mathrm{Sym}(S)$-action support the carrier-level compact matrix package

$$SU(3) \times SU(2) \times U(1),$$

where $SU(3)$ is the compact form associated with the $\mathfrak{sl}(3)$ matrix package, $SU(2)$ is the double cover of the compact $SO(3)$ branch, and $U(1)$ is the compact central factor associated with kI.

Proof. The matrix-unit carrier gives the ordinary matrix algebra Mat_3, hence the compact form attached to $\mathfrak{sl}(3)$ and the compact central circle attached to kI. The compact branch $\mathfrak{so}(3)$ is stable under simultaneous permutation conjugation, and its double cover supplies the $SU(2)$ factor. The bridge audit `l3_companion_symplectic_summary.csv` checks the compact type and carrier-level summary. □

20 Signed symplectic bridge

This bridge uses the finite pure C/J directed-edge geometry selected in Part II. The continuous statement below is a signed symplectic realization of that six-edge geometry [22].

Let $V = \mathbb{R}^6$ have ordered basis

$$\mathcal{B} = (e_{01}, e_{12}, e_{20}, e_{10}, e_{02}, e_{21}),$$

where e_{rc} denotes the directed edge (r, c). Let

$$\Omega = \begin{pmatrix} 0 & I_3 \\ -I_3 & 0 \end{pmatrix}$$

be the matrix of the symplectic form ω in this basis. Set

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

The continuation lift, unsigned reversal, sign twist, and signed reversal are

$$\widehat{C} = \begin{pmatrix} P & 0 \\ 0 & P \end{pmatrix}, \qquad \widehat{J}_0 = \begin{pmatrix} 0 & R \\ R & 0 \end{pmatrix},$$

and

$$S = \begin{pmatrix} -I_3 & 0 \\ 0 & I_3 \end{pmatrix}, \qquad \widehat{J}_s = \widehat{J}_0 S = \begin{pmatrix} 0 & R \\ -R & 0 \end{pmatrix}.$$

Here $\widehat{J}_0$ is the unsigned reversal on directed edges, while $\widehat{J}_s$ induces the same reversal on projective directed-edge lines but carries the sign needed for symplecticity.

A *signed projective dihedral lift* of the finite

$$D_3 = \langle C, J \mid C^3 = J^2 = 1,\ JCJ = C^{-1} \rangle$$

action is a pair of symplectic matrices $\widehat{C}, \widehat{J}_s \in \mathrm{Sp}(V, \omega)$ whose projective classes define a homomorphism

$$D_3 \longrightarrow \mathrm{PSp}(V, \omega),$$

together with an unsigned lift $\widehat{J}_0 \in GL(V)$ inducing the same reversal on the six directed-edge lines and satisfying

$$\widehat{J}_0^T \Omega \widehat{J}_0 = -\Omega.$$

Thus the word "signed" refers to the replacement of the unsigned anti-symplectic reversal $\widehat{J}_0$ by the signed reversal $\widehat{J}_s$, while "projective" records the relation $\widehat{J}_s^2 = -I$, which becomes $J^2 = 1$ after passing to $\mathrm{PSp}(V, \omega)$.

Theorem 20.1. The finite pure $\{C, J\}$ directed-edge geometry admits the signed projective dihedral lift displayed above. The unsigned reversal is anti-symplectic, while the signed reversal is symplectic. The finite tick word

$$C, J, C, C, J$$

visits all six directed edges and has a signed symplectic realization.

Proof. Direct matrix multiplication gives

$$\widehat{C}^T \Omega \widehat{C} = \Omega, \qquad \widehat{J}_0^T \Omega \widehat{J}_0 = -\Omega, \qquad \widehat{J}_s^T \Omega \widehat{J}_s = \Omega.$$

Moreover

$$\widehat{C}^3 = I, \qquad \widehat{J}_s^2 = -I, \qquad \widehat{J}_s \widehat{C} \widehat{J}_s^{-1} = \widehat{C}^{-1}.$$

Therefore the projective classes satisfy the dihedral relations in $\mathrm{PSp}(V, \omega)$. On the six coordinate lines, the action of $\widehat{C}$ is

$$(01\ 12\ 20)(10\ 02\ 21),$$

which is the finite continuation map $C(r, c) = (c, \overline{rc})$, and the projective action of $\widehat{J}_s$ is the reversal

$$01 \leftrightarrow 10, \qquad 12 \leftrightarrow 21, \qquad 20 \leftrightarrow 02.$$

Starting from 01, the tick word C, J, C, C, J gives

$$01 \longmapsto 12 \longmapsto 21 \longmapsto 10 \longmapsto 02 \longmapsto 20,$$

so it visits all six directed edges. The bridge audits `l3_signed_lift_matrices.csv`, `l3_companion_symplectic_summary.csv`, and `l3_tick_map_audit.csv` record the corresponding finite checks. □

Part IV

Certificates and open bridges

21 Finite certificate protocol

The support package is available at

`https://github.com/burundai-t/sigma-equivariant-magmas-ag23`.

It separates three roles:

`scripts/`, `tables/`, `mathcal_H/`.

The compact `scripts/` and `tables/` corpus supports the manuscript-facing Landscape/Selection/Linearization finite checks. The separate `mathcal_H/`

bundle supports the global hidden-continuation theorem of Section 7, and `mathcal_H/h2/verify_h2_coverage_light.py` provides the static coverage check for the accepted H2 raw artifacts.

The compact check is

```
python3 -S scripts/verify_preprint_support.py
```

It runs inventory and hygiene checks and then V1–V5:

verifier	role
V1	associativity distribution, endpoints, controlled $\mathcal{H}$ comparisons
V2	selection chain, local path-dependence, pure C/J separation
V3	operator passport, composition envelope, absorption, T_2
V4	finite bridges: C/J, $\mathcal{H}$, matrix, signed symplectic
V5	auxiliary five-dimensional fiber algebra

The compact table families are:

family	tables	purpose
L1 landscape atlas	T01–T07	associativity distribution, endpoints, controlled $\mathcal{H}$ comparison
L2 selection	T08–T11	selection chain and pure C/J survivor separation
L3 operator/term/bridge	T12–T30	operator passport, T_2, matrix bridge, signed symplectic lift, fiber algebra

The complete file-by-file index, including H2 light coverage support, is `SUPPORT_INDEX.txt`.

21.1 Global hidden-continuation certificate handles (Stage 6 bundle)

The global hidden-continuation theorem is checked by the separate bundle

```
cd mathcal_H
PYTHONDONTWRITEBYTECODE=1 python3 verify_stage6_bundle.py
```

The manuscript-level implication is

$$\text{local distance reduction} + \text{H1/H2/H3/H4 certificates} \Longrightarrow \text{Theorem 7.2.}$$

handle	bundle location	theorem component
S6-RED, S6-CERT-IF	`stage6.md`, verifier	local distance reduction
S6-H1	`h1/`	unrestricted full-landscape range
S6-H2	`h2/`	pure-frontier UNSAT certificate
S6-H3	embedded verifier check	PAB/row-complement pure-frontier witnesses
S6-H4	`h4/`	signed-cancellation spike classifier

Focused checks are:

```
PYTHONDONTWRITEBYTECODE=1 python3 verify_stage6_bundle.py --only-h1
PYTHONDONTWRITEBYTECODE=1 python3 verify_stage6_bundle.py --only-certpack-final
PYTHONDONTWRITEBYTECODE=1 python3 verify_stage6_bundle.py --only-h3
PYTHONDONTWRITEBYTECODE=1 python3 verify_stage6_bundle.py --only-h4
```

The H2 obligation proves

$$(h_q \geq 0 \ \forall q \in S^5) \wedge rawH \geq 1171$$

UNSAT on the depth-9 live frontier:

$$18540/18540 \text{ UNSAT}, \qquad SAT = 0, \qquad UNKNOWN = 0.$$

The light static H2 coverage verifier checks the accepted raw H2 artifact coverage and integrity for the declared live frontier:

```
cd mathcal_H/h2
PYTHONDONTWRITEBYTECODE=1 python3 verify_h2_coverage_light.py
```

21.2 Dependency summary

result family	support role
compact PAB uniqueness, Theorem 12.3	hand finite proof; L2 audit tables
explanatory selection chain, Proposition 12.4	didactic chain, logically subsumed by Theorem 12.3; local audits
global associativity atlas	finite atlas tables T01–T05
global hidden-continuation frontier	separate H1–H4 certificate bundle
operator passport	analytic/classification results with L3 audits
binary term algebra	sector reduction plus realization/transition certificates
matrix/Lie and signed symplectic structures	bridge audits

22 Open problems and bridges

The following fronts are subsequent work. The selection theorem and the operator passport above are complete at the level stated in this manuscript.

22.1 Closures

22.1.1 Functorial PAB–$\mathfrak{gl}(3)$ bridge

The carrier identification

$$k[M] \cong \mathrm{Mat}_3(k)$$

supports two bilinear operations: PAB multiplication and ordinary matrix multiplication. The cross-row contraction

$$\mu_{cross}(A, B) = AJ_3B^\top$$

and the carrier-level compact package give partial alignments. The closure question is whether there is a universal or functorial construction from the PAB magma to the matrix/Lie package, beyond the shared carrier.

22.1.2 Symplectic development of pure C/J

The finite pure C/J geometry has an explicit signed symplectic lift. The next problem is to develop the associated continuous symplectic geometry, including canonical (q, p)-charts, Hamiltonian paths, and coordinate-free separation of pure reversal from drifted reversal.

22.1.3 Unified selection principle

Theorem 12.3 selects PAB by the conjunction of four finite selectors of distinct nature: information-theoretic, entropic, algebraic, and combinatorial-geometric. It remains open whether these selectors arise from a single natural, non-tautological information-theoretic or algebraic functional whose global minimizers on Ω' are exactly the PAB representative. In such a formulation, PAB would appear as a global extremum of one canonical object rather than as the intersection of four independently stated extrema.

22.2 Extensions

22.2.1 All-arity term algebra

The binary term algebra satisfies

$$|T_2(\mathrm{PAB})| = 630, \qquad T_2(\mathrm{PAB}) \cong \{x, y\} \times \mathcal{S}_{21} \times \mathcal{U}_{15}.$$

The all-arity term algebra remains open: the main question is whether the binary factorization extends to a general clone structure or is genuinely binary [24, 28, 20].

22.2.2 Larger carriers

Natural generalizations include analogues over larger carriers $S \times S$, especially finite fields or cyclic sets, and directed-edge C/J geometries beyond the triangle.

22.2.3 Removing AX2

The landscape Ω' fixes the same-row Steiner axiom. The larger landscape without AX2 asks whether the same-row Steiner rule can itself be selected by intrinsic criteria.

22.3 Status summary

Front	Status
Global hidden-continuation certificate	closed by Theorem 7.2 and the `mathcal_H/` certificate bundle
Functorial PAB–$\mathfrak{gl}(3)$ bridge	open
Unified selection principle	open
Symplectic development of pure C/J	open
All-arity term algebra	open
Larger carriers	open
Removing AX2	open

23 Conclusion

The PAB magma is selected from the finite normal-form landscape $\Omega' \cong S^{21}$ by four internal finite criteria:

$$\mathcal{H}_{acc} = 0, \qquad H_{diag} = 0, \qquad \text{diagonal idempotence}, \qquad \text{pure } C/J.$$

The surrounding landscape is mapped by certified associativity and hidden-continuation atlases, and the selected magma then exposes its linear/operator passport, binary term algebra, carrier-level compact Lie package, and signed symplectic bridge.

Thus the theorem package has the intended three-layer form:

$$\boxed{\text{finite landscape}} \qquad \boxed{\text{intrinsic finite selection}} \qquad \boxed{\text{linearization of PAB}}.$$

Note on AI assistance

This work was developed with the assistance of large language models.

The author specified the conceptual framework, constraints, and target properties of the objects under study. Within this framework, AI tools were used for computational exploration, code drafting, and structuring intermediate results.

All mathematical definitions, statements, and theorems presented in the paper were established and independently verified by the author. The finite claims are backed by exhaustive checks recorded in the accompanying support package.

References

[1] K. Appel and W. Haken, "Every planar map is four colorable. Part I: Discharging," *Illinois Journal of Mathematics* **21** (1977), no. 3, 429–490. doi:10.1215/ijm/1256049011.

[2] K. Appel, W. Haken, and J. Koch, "Every planar map is four colorable. Part II: Reducibility," *Illinois Journal of Mathematics* **21** (1977), no. 3, 491–567. doi:10.1215/ijm/1256049012.

[3] C. H. Bennett, "Logical reversibility of computation," *IBM Journal of Research and Development* **17** (1973), no. 6, 525–532. doi:10.1147/rd.176.0525.

[4] C. H. Bennett, "The thermodynamics of computation—a review," *International Journal of Theoretical Physics* **21** (1982), no. 12, 905–940. doi:10.1007/BF02084158.

[5] G. Birkhoff, "On the structure of abstract algebras," *Proceedings of the Cambridge Philosophical Society* **31** (1935), no. 4, 433–454. doi:10.1017/S0305004100013463.

[6] R. H. Bruck, *A Survey of Binary Systems*, Ergebnisse der Mathematik und ihrer Grenzgebiete, Neue Folge, Heft 20, Springer, Berlin, 1958.

[7] S. Burris and H. P. Sankappanavar, *A Course in Universal Algebra*, Graduate Texts in Mathematics 78, Springer-Verlag, New York, 1981. doi:10.1007/978-1-4613-8130-3.

[8] P. M. Cohn, *Universal Algebra*, Harper's Series in Modern Mathematics, Harper & Row, New York, 1965.

[9] C. J. Colbourn and J. H. Dinitz, eds., *Handbook of Combinatorial Designs*, 2nd ed., Chapman & Hall/CRC, Boca Raton, 2007.

[10] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley-Interscience, Hoboken, 2006. doi:10.1002/047174882X.

[11] J. Dénes and A. D. Keedwell, *Latin Squares and Their Applications*, Academic Press, New York, 1974.

[12] W. Fulton and J. Harris, *Representation Theory: A First Course*, Graduate Texts in Mathematics 129, Springer-Verlag, New York, 1991. doi:10.1007/978-1-4612-0979-9.

[13] G. Gonthier, "Formal proof—the four-color theorem," *Notices of the American Mathematical Society* **55** (2008), no. 11, 1382–1393.

[14] V. Guillemin and S. Sternberg, *Symplectic Techniques in Physics*, Cambridge University Press, Cambridge, 1990.

[15] T. C. Hales, "A proof of the Kepler conjecture," *Annals of Mathematics* **162** (2005), no. 3, 1065–1185. doi:10.4007/annals.2005.162.1065.

[16] T. Hales, M. Adams, G. Bauer, T. D. Dang, J. Harrison, L. T. Hoang, C. Kaliszyk, V. Magron, S. McLaughlin, T. T. Nguyen, Q. T. Nguyen, T. Nipkow, S. Obua, J. Pleso, J. Rute, A. Solovyev, T. H. A. Ta, N. T. Tran, T. D. Trieu, J. Urban, K. Vu, and R. Zumkeller, "A formal proof of the Kepler conjecture," *Forum of Mathematics, Pi* **5** (2017), e2. doi:10.1017/fmp.2017.1.

[17] B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed., Graduate Texts in Mathematics 222, Springer, Cham, 2015. doi:10.1007/978-3-319-13467-3.

[18] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, Cambridge, 2012.

[19] R. Landauer, "Irreversibility and heat generation in the computing process," *IBM Journal of Research and Development* **5** (1961), no. 3, 183–191. doi:10.1147/rd.53.0183.

[20] D. Lau, *Function Algebras on Finite Sets: A Basic Course on Many-Valued Logic and Clone Theory*, Springer Monographs in Mathematics, Springer, Berlin, 2006. doi:10.1007/3-540-36023-9.

[21] W. McCune, "Solution of the Robbins problem," *Journal of Automated Reasoning* **19** (1997), no. 3, 263–276. doi:10.1023/A:1005843212881.

[22] D. McDuff and D. Salamon, *Introduction to Symplectic Topology*, 3rd ed., Oxford Graduate Texts in Mathematics 27, Oxford University Press, Oxford, 2017.

[23] H. O. Pflugfelder, *Quasigroups and Loops: Introduction*, Sigma Series in Pure Mathematics 7, Heldermann Verlag, Berlin, 1990.

[24] E. L. Post, *The Two-Valued Iterative Systems of Mathematical Logic*, Annals of Mathematics Studies 5, Princeton University Press, Princeton, 1941.

[25] I. G. Rosenberg, "Über die funktionale Vollständigkeit in den mehrwertigen Logiken," *Rozpravy Československé Akademie Věd, Řada matematických a přírodních věd* **80** (1970), no. 4, 3–93.

[26] C. E. Shannon, "A mathematical theory of communication," *Bell System Technical Journal* **27** (1948), 379–423 and 623–656. doi:10.1002/j.1538-7305.1948.tb01338.x; doi:10.1002/j.1538-7305.1948.tb00917.x.

[27] R. P. Stanley, *Enumerative Combinatorics, Volume 1*, 2nd ed., Cambridge Studies in Advanced Mathematics 49, Cambridge University Press, Cambridge, 2012. doi:10.1017/CBO9781139058520.

[28] Á. Szendrei, *Clones in Universal Algebra*, Séminaire de mathématiques supérieures 99, Presses de l'Université de Montréal, Montréal, 1986.